ForeDance: Structured Streaming Dance Generation with Anticipatory Music Conditioning

Abstract— **Streaming dance must prepare movements for music that has not yet arrived while continuing from an irrevocable motion prefix. This couples musical anticipation with the problem of generating coherent motion over successive decisions. We present ForeDance, a streaming dance framework that obtains forward-looking conditions from a frozen causal music model. Forecast states guide a discrete structural planner and a continuous-detail generator, providing musical lookahead without access to future audio. To make these motion components complementary, we introduce kinematics-guided structure–detail decoupling together with its training objectives: structural reconstruction preserves designated body organization and endpoint geometry, while full reconstruction retains the complete motion. Commit Forcing trains continuation from jointly committed structure and detail and the boundary state reconstructed from their decoded movement. On FineDance, forecast-conditioned models reduce a frozen music–motion matching distance relative to matched no-music training by 0.782 and 1.123 under two startup settings, with positive paired 95% intervals and consistent directions across three training seeds. These results establish that causal music forecasts provide usable conditioning within the structured streaming framework.**

I. Introduction

A dance movement often begins before its most visible accent: weight shifts prepare a step, and the body transitions toward a pose before reaching it. For a humanoid dancing to music as it arrives, these preparations must begin with incomplete knowledge of the sound they will accompany. Waiting for that music sacrifices the opportunity to prepare; using the actual future recording changes the task into offline generation. The robot must interpret the evolving music, continue coherently from its preceding movement, and realize the resulting dance through physical control. Music-driven humanoid dance therefore connects causal musical conditioning with continuous, embodied motion.

Music-to-dance models have made substantial progress in choreography and expressive human motion [1]–[4]. Audio-driven robot systems connect musical information to expressive behavior [5], and DiscoForcing [6] extends audio-driven generation to streaming control and humanoid deployment. Building on these developments, we study how a streaming generator organizes robot-native movement and continues from its own committed output while translating musical variation into appropriate dance.

This objective requires more than producing different trajectories for different soundtracks. Changes in rhythm, accents, and musical character should elicit corresponding changes in movement while preserving coherent whole-body motion. Musical suitability at the reference level does not establish suitability after execution: reference delivery and physical tracking can alter timing and attenuate motion dynamics. We therefore consider music–motion matching, response to musical conditions, and dance quality as complementary properties, distinguishing their evaluation in generated references from their evaluation in executed trajectories.

Two coupled modeling problems arise in this setting. First, a robot-native representation must organize whole-body structure while retaining local articulation. Discrete–continuous motion models provide a useful basis for this purpose [7], but combining a codebook with a residual does not by itself specify which kinematic properties each component should preserve. Second, streaming generation proceeds from an irrevocable motion prefix. Each commitment determines both the latent history and the reference state from which the next segment begins. Under Teacher Forcing (TF), these inputs come from recorded motion; at deployment, they depend on the model's own samples. Generated-history training addresses related mismatches [8]–[10], but a boundary-conditioned D+C generator must also maintain correspondence between its committed components and the state induced by their decoded movement.

We address these problems with ForeDance, a framework for causal music-conditioned generation of robot-native dance. Its kinematics-guided discrete–continuous (D+C) representation uses a shared decoder with complementary reconstruction objectives. Structural reconstruction supervises designated root, joint, and endpoint quantities, while full reconstruction supervises the complete motion. This explicitly supervises structural properties in the discrete branch while allowing a continuous residual to supply complementary detail. A causal discrete planner and a structure-conditioned residual diffusion model generate the two components.

To train continuation, we introduce Commit Forcing. A first pass samples structure and detail, commits their shared prefix, and reconstructs the next boundary state from the decoded reference motion. A second pass learns continuation under this generated history–state pair. The training context follows the same reference-update operation used during streaming inference. This state belongs to the generated reference process; measured robot state remains in the separate feedback loop of the downstream controller.

For musical conditioning, a frozen causal music model predicts upcoming music states from the audio received so far [11]. These forecasts condition both motion branches through frame-wise modulation [12], providing prospective musical context without access to the future recording. The generator repeatedly plans, commits a prefix, and updates its reference context (Fig. 1). A reference bridge connects committed robot-native motion to a fixed SONIC whole-body controller [13].

Our contributions are:

- **Kinematics-guided robot-native D+C representation and generation.** Complementary reconstruction objectives explicitly supervise structural properties in discrete codes and combine them with continuous detail through shared decoding and coupled music-conditioned generation.
- **Commit Forcing for streaming continuation.** Joint commitment of sampled structure and detail, together with reconstruction of the resulting reference boundary state, enables continuation training under a generated history–state transition.
- **A causal music-to-robot framework with layered evaluation.** Forecast-conditioned robot-native generation is connected to fixed whole-body control through a reference interface. The evaluation protocol distinguishes music matching, conditional adaptation, and motion quality before and after execution.

FineDance experiments show that forecast-conditioned models improve music–motion matching over separately trained no-music controls under both mixed-start and continuation-start training, with the same direction across three training seeds. We also examine the effect of motion-history corruption on dance quality and present a selected 60-second generation.

II. Related Work

A. Music Conditioning and Dance Generation

Music-conditioned motion depends on representations of temporal structure and musical character. Explicit acoustic descriptors extracted with librosa [14] provide rhythmic and spectral information, while pretrained models such as MERT [15] and MuQ [16] provide learned audio representations. Encoding observed music and forecasting music that has not yet arrived are distinct conditioning operations; streaming generation must specify which audio is available at each decision.

AIST++/FACT [1] and Bailando [17] develop music-conditioned choreography through sequence modeling and discrete motion representations. EDGE [2] and LODGE [18] develop diffusion-based generation and long-dance modeling, while FineDance [3] supports fine-grained full-body choreography. MEGADance [19] explores genre-aware generation, and SoulNet [4] combines hierarchical motion modeling with learned music–motion alignment. DiscoForcing [6] extends audio-driven generation to streaming character control and humanoid deployment. These developments motivate studying how musical conditions guide a robot-native generator that continues coherently from committed movement.

Generative music models offer a prospective source of conditioning. Jukebox [20] models continuation in a learned audio space, and Live Music Models [21] studies continuously generated music with synchronized control. We use a frozen causal music model, Magenta RealTime 2 [11], to predict upcoming music states from received audio. These states supply inferred musical lookahead, not access to the future recording. Our contribution concerns their integration with structured motion generation rather than a new music representation or forecaster.

B. Structured Motion and Streaming Continuation

Vector quantization [22] supports token-based motion generation in T2M-GPT [23] and MoMask [24], while latent diffusion [25] provides a continuous alternative. DisCoRD [26] connects discrete tokens with continuous motion decoding, and DC-Motion [7] combines discrete structure with continuous residuals. Our D+C design builds on this representation principle. Robot-specific reconstruction objectives explicitly supervise designated structural properties in the discrete branch, while full-motion reconstruction trains the joint representation through a shared decoder. This encourages complementary roles rather than assuming that a discrete–continuous split automatically produces kinematic specialization.

Streaming generation also requires learning to continue from generated context. Scheduled Sampling [8], Professor Forcing [27], and Diffusion Forcing [9] address training–generation discrepancies through different treatments of sequence inputs and dynamics. MotionStream [28] develops causal motion tokenization, while MotionStreamer [10] combines diffusion-based autoregression with Two-Forward training. Commit Forcing addresses the transition induced by our boundary-conditioned D+C generator: sampled structure and detail enter the committed history together, and their decoded prefix determines the next reference boundary state. Continuation is trained under this generated history–state pair, using the reference-update operation employed at inference.

C. Humanoid Motion Generation and Execution

Motion generation and physical control address complementary aspects of embodied behavior. PhysDiff [29] incorporates simulator-based correction into motion generation, while PHC [30] tracks reference motion with recovery in simulated avatars. ExBody2 [31], BeyondMimic [32], and SONIC [13] advance expressive and versatile humanoid control. Higher-level conditions connect these capabilities to human intent and perception: HARMON [33] generates humanoid whole-body motion from language, and RoboPerform [5] combines motion content with audio-derived style. Together with DiscoForcing [6], these systems provide important precedents for condition-driven robot performance.

Reference commitment also relates to receding-horizon action generation in Diffusion Policy [34] and asynchronous execution in Real-Time Chunking [35]. The latter freezes actions scheduled for execution while inpainting the remainder of a new chunk. Such mechanisms govern how action chunks are updated and delivered; Commit Forcing instead constructs the training history and boundary state induced by a joint motion commitment.

Our focus is the kinematically supervised D+C reference generator and its continuation training. A reference bridge

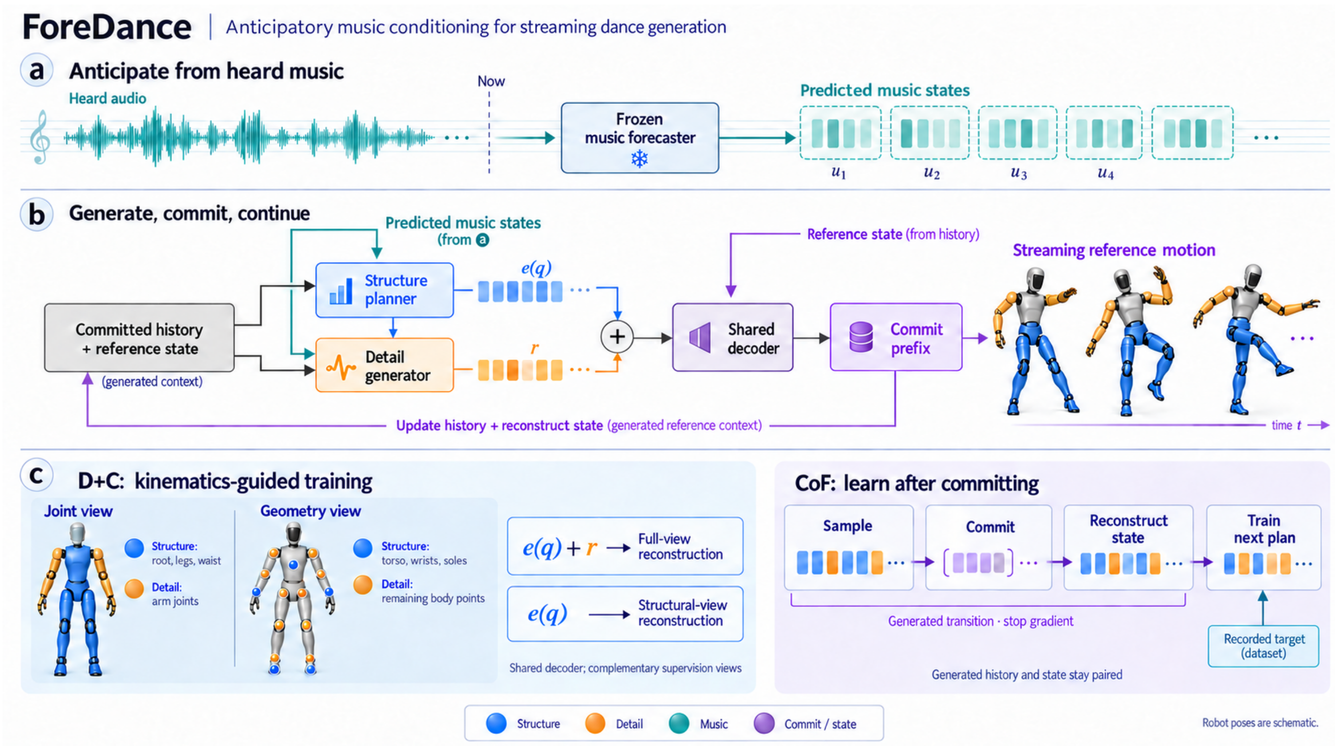


Fig. 1. **ForeDance overview. Top:** heard audio supplies a causal forecast of upcoming music states. **Middle:** the forecast conditions both discrete structure and continuous detail; structure also conditions detail. Their sum passes through a shared decoder, and the committed prefix updates the motion history and reference-derived boundary state for the next decision. **Bottom:** kinematic supervision gives the two latent components complementary roles, while Commit Forcing trains continuation from a jointly generated history–state pair. Robot poses illustrate the representation and output interface; they are schematic, not experimental or hardware execution evidence.

connects its output to a fixed downstream controller. The generator updates reference-derived context, whereas the controller closes the measured-state feedback loop. This separation motivates evaluating musical suitability and motion quality both before and after execution.

III. Problem Formulation

We consider a causal audio stream and an output motion prefix that cannot be revised. At a planning decision, let $a_{\le\tau}$ denote the arrived audio, h the committed motion-code history, and b the state derived from the last committed native frames. A frozen music model produces forecast states $m^{\text{pred}} = F(a_{\le\tau})$. The motion generator samples a plan

$$(q_{1:H}, r_{1:H}) \sim p_\theta(q, r \mid h, b, m^{\text{pred}}). \tag{1}$$

Here q is discrete structure and r is continuous detail. Both are decoded into native G1 motion; the first $C < H$ latent steps are committed and the suffix is discarded. The next decision uses the appended history and the state reconstructed from the newly committed frames.

The motion rate is 30 Hz, with two frames per latent step. We use 64 latent history steps (128 frames, approximately 4.27 s), an eight-step plan (16 frames, approximately 0.53 s), and a four-step commit (eight frames, approximately 0.27 s). The overlapping planning horizon permits revision of uncommitted motion while the output prefix grows. All inference inputs are available at the corresponding decision time.

IV. Method

ForeDance links three operations: forecasting the upcoming musical context, generating a structured motion plan, and learning to continue after committing part of that plan. Figure 1 shows how the forecast and committed motion context meet in the two motion generators.

A. Anticipatory Music Conditioning

The music that will accompany the next plan is only partly observed. We obtain a forward-looking condition by tokenizing the arrived audio with SpectroStream and rolling a frozen Magenta RealTime 2 Small model forward. The condition consists of 14 native forecast states at 25 Hz, spanning approximately 0.56 s. Each state retains its intended physical time. Forecasting in this latent music space supplies a prospective condition directly to the motion generators, while the future recording remains unobserved.

Let u_j be a forecast state and ℓ_j its lead relative to the motion decision. We form

$$c_j = W\,\mathrm{LN}(u_j) + \phi(\ell_j), \tag{2}$$

where ϕ represents physical time. The interface aligns valid forecast states with the motion horizon and uses frame-wise feature modulation [12] in both branches. In schematic form, an activation v_i receives a scale and a shift derived from its aligned condition:

$$\mathrm{FiLM}(v_i, c_i) = \gamma(c_i) \odot v_i + \beta(c_i). \tag{3}$$

Conditioning both branches lets the same musical forecast guide body organization and its complementary articulation. Timestamp-based masking excludes expired forecast states from their aligned motion positions.

The forecaster remains frozen while the dance generator learns to interpret these states. A null condition bypasses music modulation and supports separately trained no-music controls. Section V-D distinguishes the effect of using music from the additional question of whether forecast states improve on heard-music states.

B. Kinematics-Guided Structure–Detail Decoupling

The representation must retain both coordinated body motion and the articulation needed to realize it. We impose this division through kinematic reconstruction views, then use the resulting discrete and continuous codes in a coupled generator.

a) Native motion and boundary state: We represent motion directly in the robot's configuration space, including full pelvis orientation. A fixed yaw anchor R_A is removed once for a motion sequence, and every frame stores absolute orientation in that anchor:

$$R_t^A = R_A^\top R_t^{\text{world}}, \tag{4}$$

$$x_t = [\Delta p_{xy,t}^{\text{local}},\ h_t,\ \rho(R_t^A),\ \theta_t]. \tag{5}$$

Here $\Delta p_{xy,t}^{\text{local}}$ denotes planar displacement, h_t is pelvis height, ρ is the six-dimensional rotation representation [36], and θ_t contains 29 joint angles, giving 38 motion dimensions. This retains roll and pitch without multiplying a sequence of predicted rotation increments. Planar displacements are integrated using the appropriate preceding heading. The anchor remains fixed over the generated sequence; it is not reset at every commit.

The corresponding 71-dimensional boundary state is

$$b_t = [h_t, v_{xy,t}^{\text{local}}, v_{z,t}, \rho(R_t^A), \omega_t, \theta_t, \dot{\theta}_t]. \tag{6}$$

The state contains pelvis height, local planar and vertical velocities, orientation, angular velocity ω_t, joint angles, and joint velocities. Conditioning on these quantities makes the starting pose and motion explicit at each segment boundary. They are reconstructed from the committed reference.

b) Complementary latent roles: A boundary-conditioned encoder maps motion to a 16-dimensional latent per step:

$$z_i = e(q_i) + r_i, \qquad q_i \in \{1, \ldots, 512\}, \quad r_i \in \mathbb{R}^{16}. \tag{7}$$

Here $e(q_i)$ is the selected codebook embedding. A shared decoder $D_\psi(z, b)$ maps their sum to native motion. The discrete code supplies the structural hypothesis, and the residual adjusts the motion within that hypothesis. Their complementary roles are encouraged by the reconstruction objectives below; both contribute through the same decoder.

c) Training the division of labor: Let Φ_f contain native motion, boundary-aware motion velocities, forward-kinematics positions, and their velocities, each standardized using training data. In the native-motion and velocity blocks, the structural view Φ_s selects root quantities, both legs, and the waist; the complementary detail view contains the 14 arm-joint coordinates. In the geometry blocks, structure selects the torso, both wrist endpoints, and both sole points, while detail contains the remaining body points. Positions and their velocities use the same point partition. This couples body organization to endpoint placement: for example, wrist position is structural even though the corresponding arm-joint articulation lies in the complementary native view. The detail view is the exact complement within each block. Equal block weighting avoids letting a high-dimensional geometry block dominate.

Let sg denote stop-gradient. For encoder output z, embedding e, and $r = z - \text{sg}(e)$, we form full and structure-only reconstructions, together with an auxiliary residual-corrupted reconstruction:

$$x_f = D_\psi(\text{sg}(e) + r, b), \tag{8}$$

$$x_s = D_\psi(z + \text{sg}(e - z), b), \tag{9}$$

$$x_c = D_\psi(\text{sg}(e) + \mathscr{C}(r), b). \tag{10}$$

The full branch is supervised on Φ_f; the other two are supervised on Φ_s. The corruption $\mathscr{C}$ either zeros detail or adds small channel-scaled noise, encouraging the structural reconstruction to persist under changes in the residual. Figure 2 shows the two principal reconstruction paths. The combined objective is

$$\begin{aligned}\mathscr{L}_{\text{codec}} = \mathscr{L}_f + \mathscr{L}_s + 0.25\mathscr{L}_{\text{robust}} + \mathscr{L}_{\text{VQ}} \\ + \mathscr{L}_{\text{boundary}}.\end{aligned} \tag{11}$$

The final term supervises boundary-transition derivatives and pelvis orientation at the beginning of the reconstructed segment. Appendix I provides the training details.

d) Coupled generation: A causal Transformer samples structural tokens autoregressively. A continuous diffusion generator [37] conditions on the sampled structure, history, state, and forecast music to predict residual detail. We use ten DDIM steps [38]. The structural planner uses a classification objective, and the detail generator uses a denoising objective. Conditioning detail on the sampled structure connects their decisions before decoding. Committing the same prefix of both branches preserves that pairing across successive plans.

C. Commit Forcing

A committed motion segment changes two inputs to the next decision: the latent history and the boundary state. Under TF, both are taken from recorded motion; under streaming inference, both depend on the model's samples. CoF addresses this mismatch by constructing the generated transition before supervising continuation (Fig. 3). A stop-gradient first pass samples $\tilde{q}, \tilde{r}$ using the inference samplers. Their first C steps are committed together and decoded; an update operator U appends the codes and derives the new state:

$$(\tilde{h}^+, \tilde{b}^+) = U\big(h, b, \tilde{q}_{1:C}, \tilde{r}_{1:C}\big). \tag{12}$$

The next supervised plan receives this complete context. Substituting the whole transition avoids pairing generated codes with a boundary state taken from a different recorded transition.

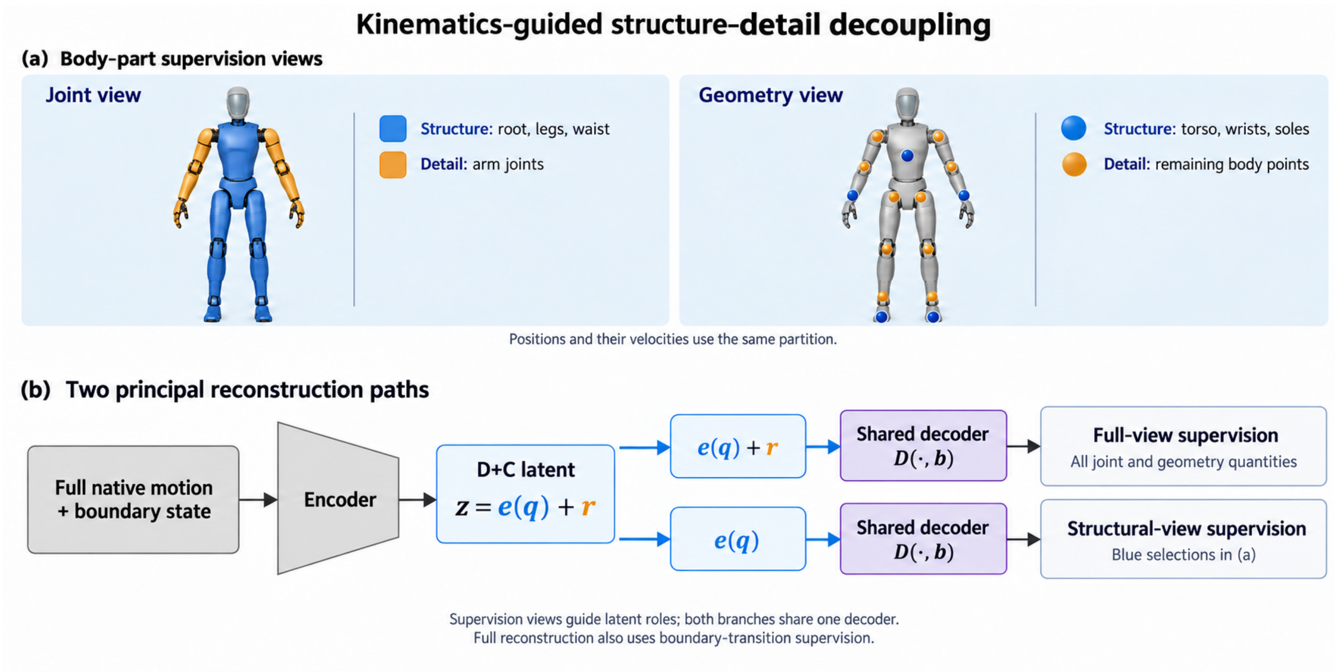


Fig. 2. **The representation and its training are one design.** Body-part supervision has separate joint and geometry partitions: root/legs/waist versus arms in joint space, and torso/wrists/soles versus other body points in geometry. The two principal reconstruction paths supervise the full latent on all views and discrete structure on the structural view. The shared decoder and boundary state are used throughout.

a) Residual targets under sampled structure: For the next dataset target $z^* = e(q^*) + r^*$ and sampled structural token $\hat{q}$, define

$$r^*_{\text{rebased}} = z^* - e(\hat{q}). \tag{13}$$

Thus $e(\hat{q}) + r^*_{\text{rebased}} = z^*$. This preserves the *target latent*; the boundary-conditioned decoder need not produce identical motion under recorded and generated states. We retain the next recorded latent target while changing the conditioning history and state. Each training example constructs its first-pass transition independently; Appendix I describes this sampling procedure.

The key operation is the joint transition: generated structure, generated detail, and reconstructed boundary state enter the next training context together. This exposes the continuation model to the consequences of its own commitment operation. Boundary continuity and long-rollout quality provide the corresponding tests against TF.

b) Supporting training choices: Full-history corruption (FHC) perturbs the complete discrete-embedding and residual history during supervised training, while leaving boundary state and forecast music unchanged. It is applied after CoF context construction and is absent at inference. This regularizes the history input across the complete context window. We consider two startup settings—mixed song-origin starts and ordinary continuation—and report their effects separately in the experiments.

D. Robot Reference Integration and Execution

Figure 4 separates reference delivery from controller feedback. The committed native motion provides the reference interface to a separate, pretrained SONIC whole-body controller [13]. A reference bridge reconstructs the root trajectory from the native representation in Eq. (5), maps joint references into the controller convention, and resamples the 30 Hz motion onto a continuous 50 Hz grid. For the current representation, planar displacement is integrated using the preceding heading, while pelvis orientation is recovered from the absolute rotation within the fixed sequence yaw anchor. The bridge does not accumulate a sequence of predicted yaw increments. Only committed motion is delivered; the uncommitted suffix is discarded before the next plan and is not sent to the controller.

SONIC remains fixed and combines the consumed reference with measured robot state to produce control actions. The generator separately updates its latent history and boundary state from committed reference motion. Appendix IV specifies the delivery and measurement protocol.

V. Experiments

We assess music–motion matching, conditional adaptation, and dance quality. The evaluation combines generation results, matched method ablations, and reference-to-execution measurements.

A. Experimental Setup

We use FineDance motion retargeted to G1 with the representation in Eq. (5). The quantitative cohort comprises 18 official test tracks, three training seeds per configuration, and three sampling seeds per trained model. All methods use the same source-supported 23 s segment beginning at source frame 264. A selected 60 s rollout provides a separate qualitative example.

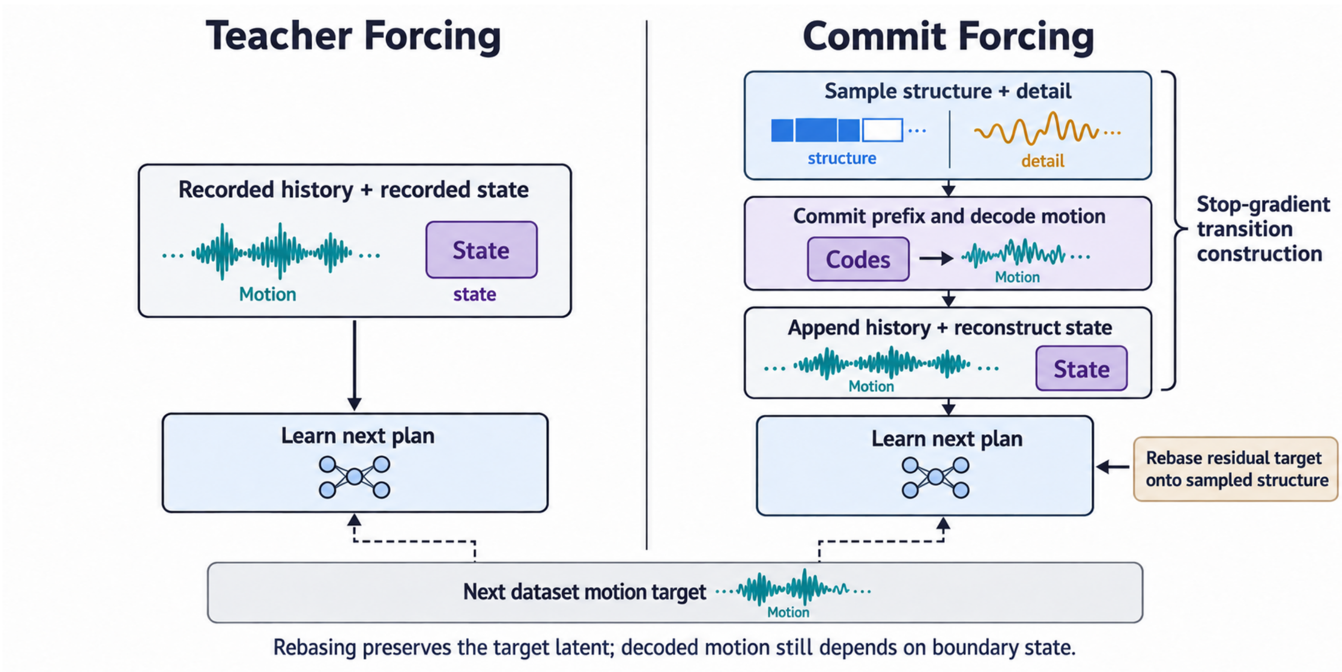


Fig. 3. **Learning the next plan: TF versus CoF.** TF receives a recorded history–state pair. CoF samples and commits structure and detail together, derives the corresponding next reference state, and supervises continuation. Both use recorded targets. Residual rebasing retains the target latent under sampled structure; decoding additionally depends on the supplied boundary state.

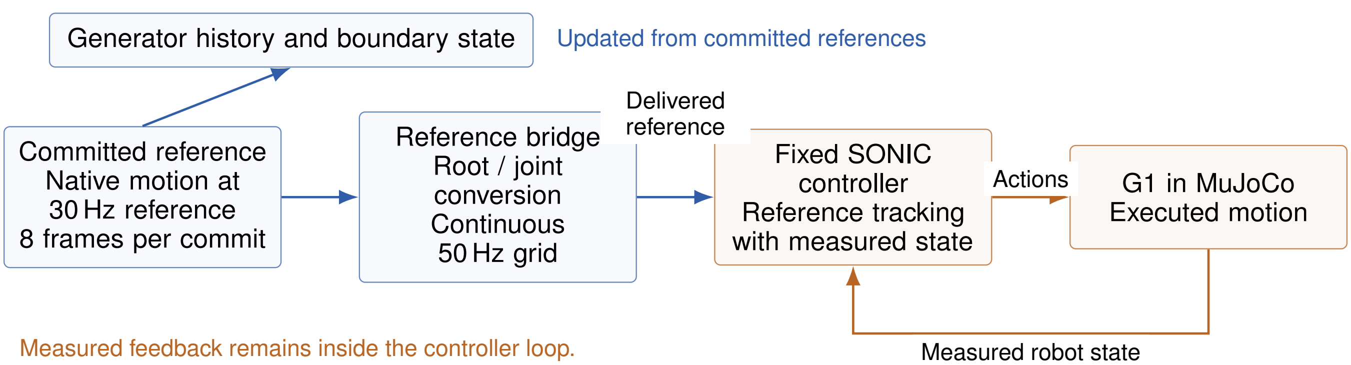


Fig. 4. **From committed references to motion tracking.** The bridge converts the reference stream onto the controller grid. Committed references update the generator context, whereas measured robot state closes the SONIC tracking loop.

Training uses 6,250 updates with batch size 48. Mixed-start training draws 20% song-origin starts and 80% ordinary continuation windows; continuation-start training uses ordinary windows throughout. The model and sampler follow Section IV-B; additional optimization details are in Appendix I. Comparisons hold source clips, initialization, and sampling settings fixed. The downstream SONIC controller is fixed.

B. Evaluation Protocol

a) Trajectory sources: The protocol distinguishes retargeted references, generated motion, and their executed counterparts. Generated motion concatenates committed prefixes. All paired scores use the same soundtrack interval and preprocessing; execution comparisons additionally retain initialization and reference-preview settings.

b) Music–motion matching: A frozen G1 audio–motion evaluator, trained independently of the generator, maps paired music and motion to a shared space. Following the alignment motivation of SoulNet [4], MMR-MS combines sequence-level correspondence and temporal feature variation; lower is better. Appendix II-A describes the evaluator construction and calibration.

c) Dance quality: Kinetic and geometric Fréchet distances (FID_k, FID_g) compare G1 motion features with retargeted reference distributions. Diversity (Div_k, Div_g) uses pairwise distances in the same feature spaces. Feature extraction, normalization, and reference collections are fixed within each comparison. Joint jerk, activity, and root/contact proxies complement distribution scores.

TABLE I

Music-condition use. Reduction in MMR-MS relative to matched no-music training; positive is favorable. Paired hierarchical 95% intervals use the complete 18-track cohort.

Startup	Improvement	95% interval
Mixed-start	0.782	[0.294, 1.285]
Continuation-start	1.123	[0.621, 1.649]

d) Rhythm and adaptation: Beat Alignment Score (BAS) measures the proximity of motion accents to music beats. Event coverage and onset–motion correlation complement rhythmic alignment. Controlled music interventions test whether generated movement responds appropriately to the specified soundtrack. Primary music scores retain the audio timeline; lag alignment is used only for tracking diagnostics.

e) Statistics: We use the complete test cohort and 10,000 hierarchical bootstrap draws, resampling training seeds and then tracks after averaging sampling repeats within each track. FID is evaluated on feature collections rather than per-song scores. Appendix II details aggregation.

C. Music–Motion Matching and Dance Quality

Table V reports matching, distribution, diversity, and contact measures for clean-history and full-history-corruption (FHC) training. Under mixed-start training, FHC reduces kinetic/geometric FID from 208.599/0.482 to 162.831/0.410, with favorable directions across all training seeds. Continuation-start changes are smaller, from 156.218/0.372 to 154.180/0.361. The effect of history regularization therefore depends on startup.

The FHC-versus-clean-history MMR-MS intervals include zero under both settings, while mixed-start contact and jerk measures favor clean history. The results indicate a distribution–smoothness tradeoff rather than uniform improvement. Complete values appear in Appendix III.

a) Qualitative generation: Figure 5 shows reference poses at 10, 30, and 50 s from one selected 60 s rollout.

b) Perceptual protocol: The planned blinded comparison separates silent-video ratings of naturalness and expressiveness from audio-enabled music-fit ratings, with matched presentation and randomized method order.

D. Music Adaptation

a) Music-condition use: Table I compares forecast-conditioned models with separately trained no-music controls. Mixed-start training reduces MMR-MS by 0.782 (95% interval [0.294, 1.285]); continuation-start training reduces it by 1.123 ([0.621, 1.649]). All three training seeds agree in direction under both settings.

b) Forecast versus heard music: The matched design in Table II varies the music states while fixing representation, conditioning capacity, CoF, startup, and training budget.

c) Conditional specificity: The paired intervention protocol uses correct, exact-null, same-track wrong-time, same-genre wrong-track, cross-genre wrong-track, and ± 4 music-frame shifts (± 160 ms). Initial history, boundary state, and sampling seed are matched; each rollout then advances with its own committed history. Outputs are scored against the designated soundtrack. These inference-time interventions differ from separately trained no-music controls.

For songs a, b and generated dances D_a, D_b, the correct-pair advantage is

$$\Delta_{\text{pair}} = \frac{1}{2}\left[d(D_a,b) - d(D_a,a) + d(D_b,a) - d(D_b,b)\right], \quad (14)$$

where d is a lower-is-better matching distance. Positive values favor intended pairings; both directional effects are retained. The cross-pairing design applies to generated and executed motion; its matched results remain to be filled.

E. Ablation Studies

1) D+C Representation and Training: The representation study measures structure/detail reconstruction, branch-swap effects, and residual effective rank. The generation comparison (Table III) independently trains discrete-only, continuous-only, and D+C alternatives under matched data and optimization settings.

2) Commit Forcing versus Teacher Forcing: Table IV changes TF to CoF while fixing the codec, music conditions, startup, history corruption, training budget, and inference sampler. Evaluation uses 60 s recursive rollouts and three non-overlapping 20 s intervals. Outcomes include matching, distribution/diversity, activity, freezing, root drift, and boundary position/velocity discontinuities.

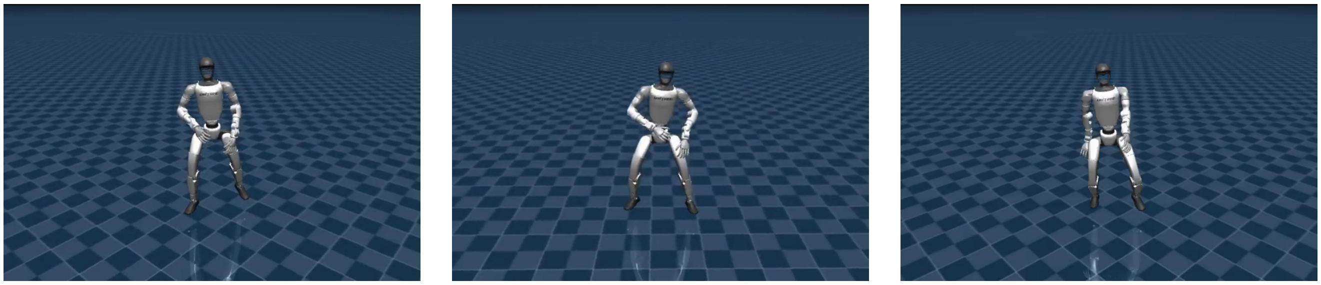
F. Online Execution and Reliability

The execution protocol compares generated, bridge-sent, controller-consumed, and measured trajectories. Joint RMSE, amplitude and squared-velocity retention measure tracking, while matching, rhythm, and motion-quality scores assess the resulting dance. Executed dataset references provide a calibration control. Supporting tracking measurements are given in Appendix IV.

Offline–online controls separately vary condition content/age and reference delivery, retaining source-trajectory identity and preview duration. Runtime reporting includes valid forecast coverage, planning-time tails, deadline misses, and stale-condition fallback. Startup and steady state are reported separately, with all attempts included in completion statistics.

VI. Discussion

a) Music conditioning and motion quality: Forecast-conditioned models improve matching over separately trained no-music controls under both startup settings. The FHC results reveal a different tradeoff: distribution distances improve most under mixed-start training, while contact and jerk favor clean history. Matching, distributional fidelity, and smoothness therefore provide complementary evidence rather than an interchangeable measure of dance quality.



10 seconds | 30 seconds | 50 seconds

ForeDance: selected 60-second FineDance generation.

Fig. 5. **A 60-second ForeDance generation.** Original frames at 10, 30, and 50 s from a selected FineDance sequence. The images show rendered reference motion.

TABLE II

FORECAST VERSUS HEARD-MUSIC CONDITIONING. MATCHED COMPARISON DESIGN; RESULTS ARE PENDING (—). EACH METRIC USES IDENTICAL SOURCES AND SAMPLING CONDITIONS ACROSS METHODS.

Condition	MMR-MS ↓	BAS ↑	FID_k ↓	FID_g ↓	Div_k	Div_g
Heard-music states	—	—	—	—	—	—
Predicted-music states	—	—	—	—	—	—

TABLE III

REPRESENTATION AND TRAINING COMPARISON. MATCHED GENERATOR RESULTS ARE PENDING (—).

Representation	FID_k ↓	FID_g ↓	Div_k	Div_g
Discrete only	–	–	–	–
Continuous only	–	–	–	–
D+C with paired training	–	–	–	–

TABLE IV

COMMIT FORCING VERSUS TEACHER FORCING. MATCHED 60 S ROLLOUT RESULTS ARE PENDING (—).

Training	FID_k ↓	FID_g ↓	Div_k	Div_g
Teacher Forcing	–	–	–	–
Commit Forcing	–	–	–	–

b) Scope and limitations: The numerical generation study covers 18 FineDance test tracks, while the 60 s sequence is a qualitative example. The no-music comparison establishes condition use, not the incremental benefit of forecasting over heard-music features. Matched D+C and TF/CoF ablations remain needed to isolate their gains. End-to-end musical adaptation and perceptual quality are not established by the current reference-generation results or supporting tracking measurements.

VII. CONCLUSION

We introduced ForeDance, combining causal music forecasts with robot-native D+C generation and Commit Forcing continuation training. FineDance results show improved music–motion matching over no-music training under two startup settings, while history regularization exposes a distribution–smoothness tradeoff. The framework connects its committed reference stream to fixed whole-body control and provides a basis for evaluating musical expression through generation and execution.

APPENDIX I
REPRESENTATION AND TRAINING DETAILS

a) Kinematic reconstruction losses: The full view contains four blocks: native motion, native velocity, forward-kinematics positions, and geometry velocity. Each block is standardized with training-split statistics. For a view Φ with blocks $\Phi^{(k)}$, define the reconstruction error

$$\mathscr{E}_{\Phi}(\hat{x},x) = \frac{1}{4}\sum_{k=1}^{4} \mathrm{mean}\left[\left(\Phi^{(k)}(\hat{x}) - \Phi^{(k)}(x)\right)^2\right]. \tag{15}$$

This gives $\mathscr{L}_f = \mathscr{E}_{\Phi_f}(x_f, x)$, $\mathscr{L}_s = \mathscr{E}_{\Phi_s}(x_s, x)$, and $\mathscr{L}_{\mathrm{robust}} = \mathscr{E}_{\Phi_s}(x_c, x)$. Equal block weights prevent the number of geometric coordinates from determining their relative contribution. The VQ term is the codebook loss plus 0.25 times the commitment loss.

For the auxiliary robustness objective, the residual is zeroed with probability 0.5. Otherwise, Gaussian noise is added with scale sampled uniformly from 0.01–0.05 times the residual channel standard deviation. The resulting reconstruction is supervised on the structural view. This codec objective is separate from FHC, which regularizes the generator's motion history.

The boundary loss supervises the first three decoded frames relative to the supplied boundary. It combines normalized smooth-L_1 errors for velocity, acceleration, and jerk with weights 1, 0.5, and 0.25, respectively, plus pelvis-rotation geodesic error with weight 1. These quantities explicitly penalize a reconstructed segment whose beginning is inconsistent with the target transition.

b) Body-part supervision views: In the native block, structure selects the first 24 of 38 coordinates: nine root coordinates, twelve leg joints, and three waist joints. The remaining 14 coordinates are the bilateral shoulder (three per arm), elbow (one), and wrist (three) joints. In the 35-dimensional velocity block, structure selects the first 21 coordinates: root linear/angular velocity and the same 15 leg/waist joint velocities. The other 14 are arm-joint velocities.

The geometry block contains 31 points: 29 non-root body origins and two lowest-foot geometry points. Structure selects the torso, the two wrist-yaw body origins, and the two lowest-foot points. Detail selects the remaining 26 points. Geometry velocity uses the same partition. These are supervision and analysis views, not independent anatomical decoders. In particular, the structural geometry view can constrain wrist placement while the complementary native view describes arm articulation.

c) Generator optimization and initialization: The update budget in Section V-A corresponds to 300,000 training windows. Residual sampling uses guidance fixed at 1.0; the music model remains frozen during generator training.

d) History regularization: FHC applies to the complete 64-step history with probability 0.5 per example. The discrete embeddings and continuous residual history share a severity timestep sampled from 850–999, with independent Gaussian noise. Their masks and timestamps are preserved, as are the boundary state and forecast conditions. Corruption is applied in the supervised pass after CoF context construction and is disabled at inference.

e) Constructing CoF transitions: Figure 6 details the construction. Each training example independently samples a first-pass transition from its supplied context. The committed discrete and residual prefixes enter the history together, and their decoded motion determines the next boundary state. The next recorded structural and latent targets remain fixed. Residual rebasing expresses that latent target relative to the sampled structure; it does not re-encode the target under the generated boundary. This construction trains local continuation after a generated commit without requiring a fully recursive long training rollout.

APPENDIX II
EVALUATION PROTOCOL

The cohort in Section V-A yields 162 track-level outputs per configuration. For each training seed and track, sampling repeats are averaged before computing paired method differences. Bootstrap sampling retains these pairs, resampling training seeds with replacement and tracks within each sampled seed. Percentiles 2.5 and 97.5 define the reported 95% interval. Qualitative examples and unfilled comparisons do not enter these estimates.

A. Metric Construction and Version Boundaries

a) Learned evaluator construction: The archived G1 evaluation pipeline combines 35-dimensional librosa temporal audio descriptors with frozen MuQ-large global audio features [14], [16]. Audio and native G1 motion branches are fitted on paired training data with global/local contrastive objectives and temporal-misalignment controls, then frozen for scoring. Music identity, rather than dancer or performance identity, determines valid positives. These are evaluation features, not additional inputs to the causal generator.

The archived global–temporal matching definition, following SoulNet [4], is

$$D_{\text{temp}} = \sum_{t=2}^{T} \|(u_t - u_{t-1}) - (v_t - v_{t-1})\|_2, \tag{16}$$

$$d_{\text{MMR}} = \sqrt{0.7\|u_g - v_g\|_2^2 + 0.3 D_{\text{temp}}}, \tag{17}$$

where u_g, v_g are global audio/motion embeddings and u_t, v_t describe aligned, non-overlapping one-second segments. MMR-MS averages paired distances. The temporal term sums norms, not squared norms or time-averaged norms, so durations must match. A global-only distance is not this complete score. The current 23 s results are retained from their archived summaries without recomputation; binding their evaluator receipt to this exact formula, feature fusion, input channels, and normalization remains a metric-provenance check before submission.

b) Calibration: Correct-versus-wrong-song and wrong-time margins, retrieval with declared candidate pools, and controlled motion degradation test metric sensitivity. Same-song multiple performances are valid positives, not false negatives. The earlier FineDance+AIST++ candidate was calibrated on 28 independent songs (18 FineDance and ten AIST++). Aggregate and FineDance gates passed, but several AIST++ wrong-song, same-style, and same-tempo confidence intervals included zero. It is not accepted for cross-dataset model ranking. This calibration is separate from generation generalization, and neither establishes evaluator validity on executed trajectories. The final evaluator training, calibration, and test-song manifests must make any reuse explicit.

c) G1 distribution and local-motion features: Kinetic descriptors summarize motion dynamics, while geometric descriptors summarize body configurations and spatial relations in the G1 domain. FID compares feature means and covariances with those of retargeted references; diversity measures pairwise distances in the same feature space. Extractor version, coordinate convention, normalization, crop length, and sample counts are part of each result. The earlier three-model table and current ForeDance results are not pooled: their numerical feature-scale equivalence has not been established. A multi-seed FID aggregation must retain how collections were formed; it is not a mean of single-song FIDs. For future execution scoring, generation and execution must use identical feature extraction. Exact feature lists and activity/contact thresholds require the corresponding implementation receipt.

d) Beat and dynamic-response measures: For motion-event times $\mathscr{B}_x$ and music-beat times $\mathscr{B}_a$, the motion-to-music BAS definition is

$$\text{BAS} = \frac{1}{|\mathscr{B}_x|} \sum_{b \in \mathscr{B}_x} \exp\left(-\frac{\min_{a \in \mathscr{B}_a} (b-a)^2}{2\sigma^2}\right). \tag{18}$$

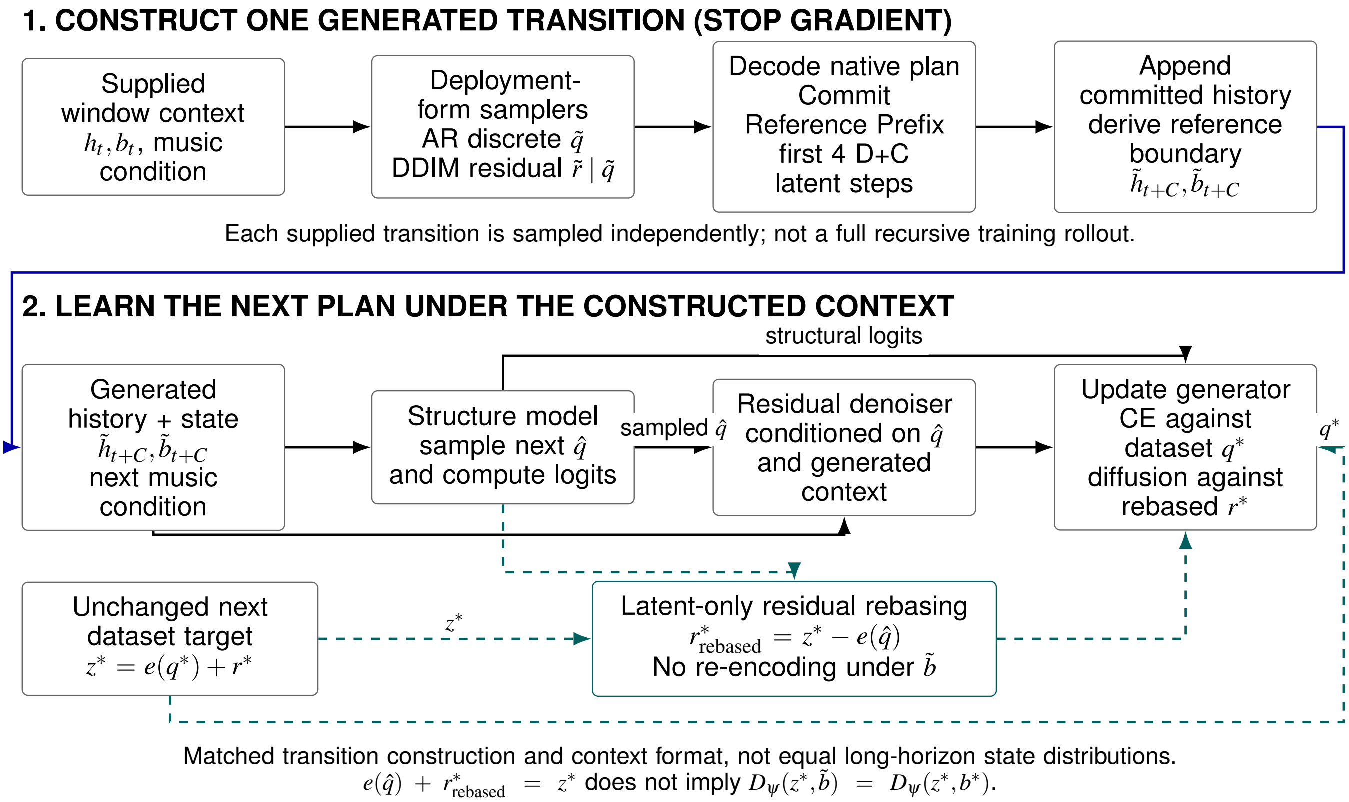


Fig. 6. **Commit Forcing transition construction and supervision.** A sampled, jointly committed structure–detail prefix supplies the next history and reference boundary. The next dataset latent remains fixed; residual rebasing expresses it relative to the sampled next structure. This preserves the latent target, while the changed boundary can change its decoded motion. The two stages are constructed independently for each supplied training example.

The earlier layer analyzer uses minima of smoothed mean joint speed with $\sigma = 0.10\,\mathrm{s}$. This does not automatically specify the extractor used for the current-generation BAS summaries, which retain their own metric version. Event counts, empty-event handling, smoothing, and any beat-matching tolerance must accompany reproducible scores. Beat precision/recall are auxiliary coverage diagnostics, not requirements that dance hit every audio beat. The layer analyzer's impact proxy is a smoothed absolute derivative of the speed curve, not contact force. Zero-lag correlation with audio onset strength measures aligned dynamic response; best-lag correlation and its lag are diagnostics, not permission to retime primary music scores.

Appendix III
History-Corruption Results

Table V provides the full values for the comparison in Section V-C.

Appendix IV
Reference Delivery and Supporting Execution Diagnostics

a) Replay setup: Four generated trajectories from one FineDance track and one sampling seed are replayed in MuJoCo for approximately 60 s each.

b) Reference construction and delivery: A four-step latent commit contains eight 30 Hz frames. A continuous 50 Hz time grid yields 13 or 14 reference samples per commit without resetting the sampling phase at each packet. The controller receives committed references rather than motor torques. The four diagnostic replays use the same fixed SONIC controller at real-time playback, with no uncommitted-plan preview, clipping, startup ramp, or alignment hold. Initial state, delivered future-reference access, and packet timestamps are part of the protocol; full-trajectory delivery and committed-prefix delivery are not interchangeable controls. The trajectories begin at an artificial continuation point with empty motion history, rather than at full-song onset.

c) Observed layers: Let M_{gen}, M_{sent}, M_{target}, and M_{exec} denote generated, resampled, consumed, and measured motion. Target and measured joints use named fields in MuJoCo order, with matched coordinate conventions, offsets, and timestamps.

d) Conditions and measurements: The first three trajectories share cached forecast content but use different age traces: ideal age, the earlier measured trace, and a revised measured trace. The fourth uses live forecast content with the revised trace. These labels identify the generation conditions of the saved trajectories.

For active joint j, amplitude and velocity-variation retention

TABLE V

EFFECT OF HISTORY CORRUPTION ON THE FULL 18-TRACK COHORT. LOWER IS FAVORABLE FOR MMR-MS, FID, AND PFC. DIVERSITY DESCRIBES MOTION SPREAD AND IS INTERPRETED TOGETHER WITH FIDELITY.

Startup	History	MMR-MS	FID_k	FID_g	Div_k	Div_g	PFC
Mixed-start	Clean history	40.902	208.599	0.482	7.495	0.493	0.143
Mixed-start	History corruption	40.859	162.831	0.410	8.626	0.508	0.167
Continuation-start	Clean history	41.195	156.218	0.372	8.329	0.545	0.163
Continuation-start	History corruption	41.029	154.180	0.361	9.017	0.541	0.162

are

$$R_j^{\mathrm{amp}} = \frac{P_{95}(q_j^{\mathrm{exec}}) - P_5(q_j^{\mathrm{exec}})}{P_{95}(q_j^{\mathrm{target}}) - P_5(q_j^{\mathrm{target}})}, \tag{19}$$

$$R_j^{\mathrm{dyn}} = \frac{\mathbb{E}[(\dot{q}_j^{\mathrm{exec}})^2]}{\mathbb{E}[(\dot{q}_j^{\mathrm{target}})^2]}. \tag{20}$$

Only joints with a target percentile range above 0.02 rad enter the active-joint summary, and ratios with near-zero denominators are excluded. The reported retention values are medians of joint-wise ratios. The dynamic ratio measures squared joint velocity; it is not physical energy or a whole-body energy ratio. Joint-position RMSE is evaluated after time alignment. Alignment lag is estimated within ± 0.5 s on the 50 Hz grid; the four recorded estimates are 40 ms, which is a case-specific alignment result rather than a general controller or system latency. Steady-state summaries exclude the first two seconds, so they do not establish startup behavior.

e) Interpretation: Amplitude retention ranges from 0.858 to 0.901, while squared-velocity retention ranges from 0.411 to 0.496. Position tracking thus coexists with dynamic attenuation. High-frequency variation can include articulation and jitter; these measurements alone do not determine the musical value of the attenuation.

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TABLE VI

SUPPORTING FIXED-REFERENCE REPLAY MEASUREMENTS. DATA FROM THE EARLIER 34D GENERATOR, SEPARATE FROM THE CURRENT 38D COHORT. EACH ROW IS ONE MUJOCO REPLAY THROUGH FIXED SONIC, EXCLUDING ITS FIRST TWO SECONDS; MUSIC INFERENCE IS NOT RUN DURING REPLAY.

Reference-generation condition	Aligned RMSE (rad)	Median R^{amp}	Median R^{dyn}
Cached forecast, ideal age	0.1421	0.869	0.496
Cached forecast, earlier age trace	0.1281	0.858	0.428
Cached forecast, revised age trace	0.1394	0.877	0.412
Live forecast, revised age trace	0.1431	0.901	0.411

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